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CSA5112 - Cryptography and Network Security [CO1 AT2]

1. CONCEPT MAPPING : Information Security

Security Attacks
Actions that exploit vulnerabilities to compromise a system or its data.

Threats
Potential conditions or events that can cause harm to a system, data or organisation

Risks
The possible negative impacts on assets, operations or objectives if a threat is realized

Security Goals
Fundamental principles that protect information and systems by reducing vulnerabilities and risks

Confidentiality
prevents unauthorized access
- Encryption
- Access control
- Authentication

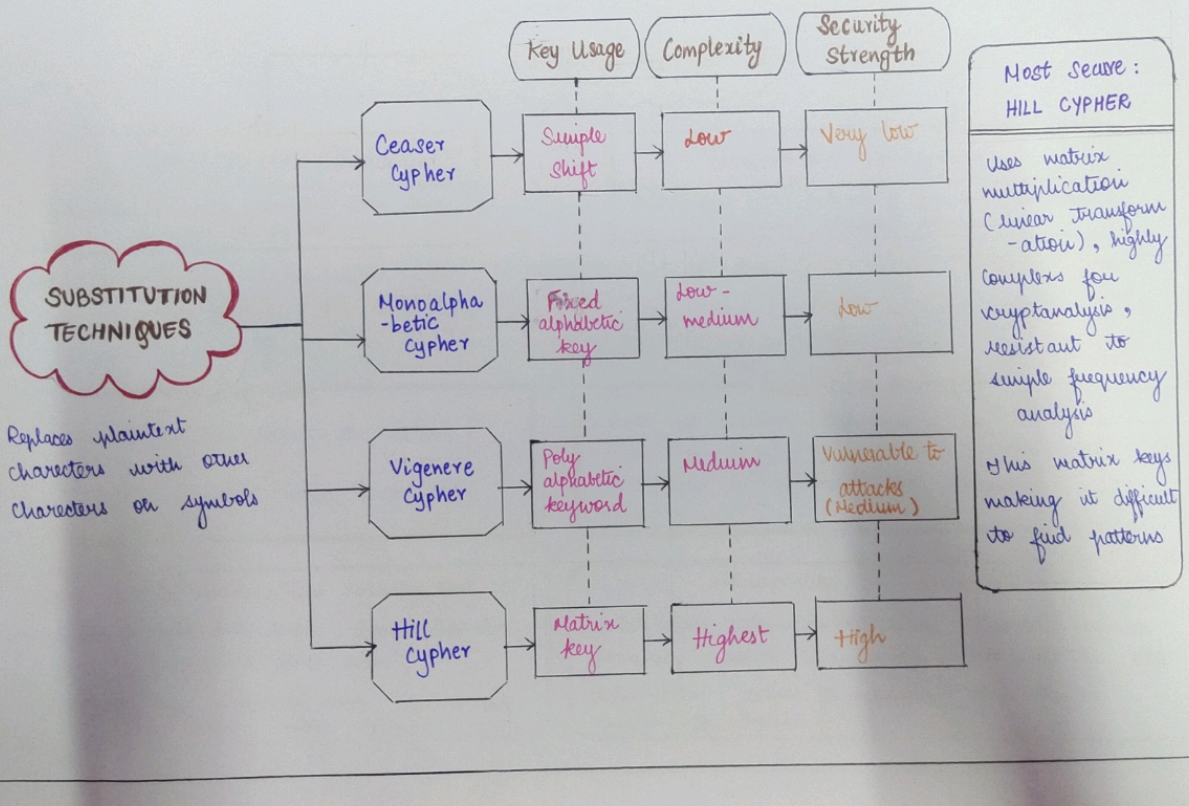
Integrity
prevents unauthorized changes
- Hashing
- Digital signatures
- Input validation

Availability
ensures access when needed
- Backup & redundancy
- Firewalls
- Disaster recovery

Real world example :

→ A phishing email (attack) tricks an employee into revealing their login credentials (threat), which leads to unauthorized access to the company's database (risk), causing data breach and financial loss.

2. Substitution Ciphers : Comparative Analysis



3. Classical Encryption Schemes



Substitution techniques

Replaces plaintext characters with other characters or symbols

eg: Caesar cipher, Monoalphabetic cipher, Vigenere cipher, Hill cipher

Transposition techniques

Rearranges the positions of plaintext characters without changing them

eg: Rail fence cipher, Columnar cipher, Route cipher

Differences

- Substitution changes the actual characters
- Transposition only changes the positions

Similarities

- Both are classical techniques
- Both use key (shift / mapping or permutation
- Both can be combined for stronger security

Effect on data structure and security level

Substitution : same data length ; changes character value → lower security

transposition : keeps characters same ; changes order → slightly higher security

Vulnerability to attacks

Substitution techniques are more vulnerable because frequency analysis can reveal patterns in the letter distribution ; while transposition requires more analysis to break.

4. Steganography Concepts

